Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/22/2020
PI Name:	Poirier, David Sc.D.		
Project Title:	Effect of Varying Convection on Dendrite Morphology and Macrosegregation		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	12/01/2014	**End Date:**	08/13/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/13/2021 per NSSC info (Ed., 6/25/21) NOTE: End date change to 4/30/2021 per NSSC information (Ed., 9/10/2020) NOTE: End date change to 4/30/2020 per NSSC information (Ed., 11/13/19)		
Key Personnel Changes/Previous PI:	October 2017 report: Prof. D.R. Poirier, Dept. of Materials Science and Engineering, The University of Arizona, remains as the P.I. September 2020: Dr. V. Manga, Assistant Research Professor, in the Dept. of Materials Science and Engineering at The University of Arizona has been the co-investigator since early 2016. He worked on the program until April, 2020. A no-cost extension was awarded so the research will continue until April, 2021. During the period of the extension, Mr. S. Kirchsmann has been hired as a Senior Programmer (part-time) to update the software that is being used to simulate the directional convection. Also Dr. M. Lauer has been hired as Senior Researcher (par-time) to effect simulations and to extend the scope of simulations to study the role of the thermal properties of the mold used for directional solidification.		
COI Name (Institution):	Manga, Venkat Ph.D. (The University of Arizona) Tewari, Surendra Ph.D. (Cleveland State University)		
Grant/Contract No.:	NNX15AC04G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	NOTE (Ed., 12/16/2014): This project is a continuation of grant NNX10AV40G, "Effect of Varying Convection on Dendrite Morphology and Macrosegregation," period of performance of 12/1/2010-11/30/2014, and with the same Principal Investigator (PI), Dr. David Poirier. The research of this grant pertains to convection that is driven by directional solidification (DS) through changes in cross-sections (because of solidification shrinkage) and thermosolutal convection. In microgravity, of course, the thermosolutal convection will be mitigated so that effects of the shrinkage-driven flow in cross-sectional changes can be isolated from the usual thermosolutal convection that always occurs when solidification occurs on Earth. The outcome of the research is expected to be relevant to commercial DS-castings, which are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and electric power generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research not only has scientific merit, it is also relevant to the production of commercial DS castings. The research is on directional solidification experiments that are compatible with the solidification cartridge assemblies (SCAs) that can be employed in a furnace on the International Space Station (ISS). The research comprises: 1) selected alloys have varying density-responses to gravity forces; 2) crucible designs with variable cross-sections but compatible with the current SCAs; 3) sophisticated characterizations of the solidified microstructures, including both micro- and macro-segregation; and 4) supporting computer simulations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The outcome of the research is expected to be relevant to commercial directionally solidified-castings, which are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and turbine electric power-generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research is very much needed for its scientific merit and also because it is relevant to the production of commercial DS castings.
Task Progress:	During this period, Dr. Manga (post-doctoral researcher) was assigned to do computer simulations of the transport phenomena and solidification during directional solidification (DS) of a binary alloy. The DS-casting process is relevant to the manufacture of critical parts in gas-turbine engines. The computer simulations of the DS in the presence of gravity and in 0-g of an Al-7Si alloy into a contraction of the cross-section were initiated but not completed. Such simulations are important in order to demonstrate the significant role of buoyancy driven convection on the DS-process. As of the this writing, the goal is to report on these findings in the Final Report and to disseminate the results to the solidification community by means of publication. In the period remaining for completion of the grant research, in addition to reporting on the Al-7Si alloy as mentioned above, we also plan to study the role of the thermal properties of the mold used for the DS. A former PhD student did simulations of the Al-7Si alloy using both alumina-molds and graphite-molds. At the time these materials were candidates for space experiments. The space experiments are no longer planned, but this is mentioned here because significant differences in the convection and the solidification segregation were seen. In the time remaining for the grant, we plan to explore different mold materials (i.e., "shell" materials) that are used in commercial practice for DS-castings. Our goal, of course, is to also disseminate these results to the solidification community.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Upadhyay SR, Tewari SN, Ghodes M, Grugel RN, Poirier DR. "Primary dendrite trunk diameter in Al-7wt% Si alloy directionally solidified aboard the International Space Station." IOP Conf Ser: Mater Sci Eng. 2019;529:012022. https://doi.org/10.1088/1757-899X/529/1/012022 , May-2019